Digital Forensic Evidence Management System

Problem Statement

Police departments, forensic labs, and courts deal with an ever-growing pile of digital evidence — seized devices, disk images, log files, memory dumps, network captures, mobile extractions, multimedia files — collected during criminal investigations. In most most agencies, this evidence is still tracked through manual logs, Excel sheets, or a mix of disconnected local systems. This approach breaks down in several specific ways:

Chain of custody gaps

There’s no single, tamper-evident record showing who collected, accessed, transferred, or analysed a given piece of evidence, and when. If custody can not be proven end-to-end, defense counsel can get evidence thrown out or ruled inadmissible.

Data integrity concerns

Without a system that records and checks hash values (MD5, SHA-256) at every stage, there's no reliable way to prove evidence hasn't been altered — intentionally or otherwise — since the moment it was collected.

Weak case-evidence linkage

Because records aren't relational, investigators struggle to connect evidence to the right case, suspect, victim, or officer, which slows investigations down.

Access control weaknesses

Sensitive evidence is frequently reachable without proper authentication or role-based restriction, and without an audit trail, which opens the door to leaks or tampering.

Poor searchability

Finding a specific item by case ID, evidence type, date, or examiner is slow and error-prone when everything lives in paper files or unindexed spreadsheets.

No lifecycle tracking

There's no dependable way to see where an item stands — collected, in storage, under analysis, presented in court, released, or disposed — so items get lost or forgotten in storage more often than anyone would like.

No standardized reporting

Producing audit reports, custody logs, or case summaries for court is a manual, time-consuming job that invites human error at exactly the point where accuracy matters most.

These aren't just administrative headaches — they're legal risk. A single gap in a custody record can get an entire piece of evidence excluded, which can be the difference between a conviction and an acquittal. The people directly affected are investigators and forensic examiners (who lose time hunting for records), evidence custodians (who have no reliable way to track lifecycle status), prosecutors and defense counsel (who need a verifiable audit trail), and ultimately the courts and the public, since the integrity of the justice system depends on evidence being handled correctly. A database-driven system that enforces custody logging, hash verification, and role-based access directly addresses all of this.

Project Objectives

Objective 1 — Design a relational database to manage the evidence lifecycle.

Build a normalized schema (3NF) that tracks evidence records through every stage — collection, storage, analysis, court presentation, and disposal.

Expected outcome: every evidence item has a traceable status at all times, bringing lost or misplaced records down to zero.

Objective 2 — Implement tamper-evident chain-of-custody tracking.

Build a custody log module that automatically records every access, transfer, or modification event, capturing who did what and when.

Expected outcome: a complete, court-admissible custody trail for 100% of evidence items, cutting manual custody-log prep time by at least 70%.

Objective 3 — Ensure data integrity through cryptographic hash verification.

Auto-generate MD5/SHA-256 hashes at intake and re-check them at every retrieval or transfer to catch tampering.

Expected outcome: 100% hash-match verification accuracy, with mismatches flagged within seconds.

Objective 4 — Enforce role-based access control (RBAC).

Only permitted roles — Admin, Investigator, Forensic Examiner, Legal Reviewer — can view, edit, or export specific evidence data.

Expected outcome: zero unauthorized access incidents, with every access attempt (successful or failed) logged and auditable.

Objective 5 — Improve search, retrieval, and reporting efficiency.

Add indexed search (by case ID, evidence type, date, examiner, status) and automated report generation for audits and court.

Expected outcome: cut average retrieval time from the current 15–30 minutes down to under 10 seconds, and automate 90% of custody/audit report generation.

Literature Survey

1. Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics

Authors: SEFCOM/ASU research group. Year: 2024.

Methodology: A broad survey of current chain-of-custody practices across academic research and real-world forensic use, sorted into three buckets — traditional paper trail, system-oriented, and infrastructure-driven approaches — and evaluated against a framework covering legal, security, and practical considerations.

Findings: The paper maps out where each approach is strong and where it falls short, and proposes directions for future work.
Limitations: It's comparative and theoretical, not an implemented system — there's a taxonomy and an evaluation framework, but no working database.

2. Digital Evidence Management Framework (DEMF)

Authors: Cosić et al. Year: 2010 (cited within the 2024 ASU survey above).
Methodology: Proposed a structured framework for tightening chain-of-custody documentation at every stage of an investigation, capturing key custody-transition data points as evidence moves through the process.
Findings: One of the earliest attempts to formalize digital chain-of-custody documentation, and it shaped a lot of the system-oriented designs that came after it.
Limitations: Being an early framework, it predates hash/blockchain-based integrity checks and leans on centralized record-keeping, which is vulnerable to insider tampering.

3. Digital Chain of Custody: An Ontology-Based Approach

Authors: ResearchGate contributors (bibliometric/ontology study). Year: 2018.
Methodology: Builds a taxonomy and an ontological model to formally represent the relationships between evidence, custodians, and case events.
Findings: The ontology gives a new lens for studying chain of custody and could feed into future standards. It also reinforces a basic legal point — for evidence to hold up in court, it has to be clearly documented who touched it, when, where, why, and how, at every step.
Limitations: The model stays conceptual — there's no empirical testing of query speed or scalability on an actual running database.

4. A Bibliometric Analysis and Systematic Review of Blockchain-Based Chain of Custody for Digital Evidence

Authors: Springer Nature (systematic review). Year: 2024.

Methodology: A systematic review and bibliometric analysis of blockchain-based chain-of-custody research, drawing on major databases from 2015 onward.
Findings: Confirms the appeal of blockchain for an immutable, decentralized audit trail, and notes growing research momentum in the space.
Limitations: As a meta-review, it doesn't test real deployment costs, latency, or storage overhead in an actual forensic lab — something the authors flag as open for future work.

5. ForensicChain: Blockchain-Based Secure Digital Forensic Investigations

Authors: Springer Nature research team. Year: 2025.
Methodology: Proposes ForensicChain, a system for securely recording, updating, and verifying case data, using Merkle trees to fingerprint case files for efficient tamper detection. A proof-of-concept was built on the Cardano blockchain.
Findings: Shows that authorized actions get reliably recorded with a fully auditable modification history, and demonstrates that Merkle-tree tamper detection works in practice.
Limitations: Proof-of-concept only — real-world scalability, transaction cost, and interoperability across agencies weren't fully tested.

6. Blockchain-Based vs. SQL Database Systems for Digital Twin Evidence Management: A Comparative Forensic Analysis

Authors: Boyd Franken, Hong-Hanh Nguyen-Le, Nhien-An Le-Khac (University College Dublin). Year: 2025.
Methodology: Ran controlled experiments comparing Ethereum + IPFS storage against a standard MySQL database, measuring storage and retrieval times for digital twin evidence between 1MB and 200MB.
Findings: Blockchain wins on data integrity and immutability, but a traditional relational database wins on performance consistency — which is directly

relevant here, since it supports the case for a well-built relational DFEMS paired with hash-based integrity checks rather than a full blockchain stack.
Limitations: Only tested on digital twin/metaverse-style data, so results may not generalize cleanly to things like mobile extractions or memory dumps.

7. Interoperability Between Digital Evidence Management Systems (Patent/System Design)

Authors: Industry patent filing (USPTO). Year: Filed publication.
Methodology: Describes an architecture where an evidence database (fed by capture devices like body cameras) is paired with a separate chain-of-custody database that logs the chronological sequence of control, transfer, copying, analysis, and disposition.
Findings: Demonstrates a practical dual-database design that supports data exchange between different agencies' systems.
Limitations: It's a patent filing, not peer-reviewed research — no independent performance or security validation, and the focus is interoperability rather than access control or search efficiency.

Analysis of the Existing System

What the current (manual/spreadsheet-based) approach gets right:

- It's simple and familiar — no technical training needed, so non-technical staff can pick it up immediately.
- Low upfront cost — no licensing, servers, or database administration required.
- Paper chain-of-custody forms already have long-standing legal acceptance in most jurisdictions.
- Works offline, since it doesn't depend on network connectivity or system uptime.
- Spreadsheets are easy to reshape for a new field or an unusual case requirement.

Where it falls short:

- **No centralized storage** — records for one case can be scattered across paper files, spreadsheets, and personal drives, so retrieval is slow and inconsistent.
- **No real-time integrity checks** — hash values, when recorded at all, are usually noted by hand and never re-verified, so tampering can slip through unnoticed.
- **Weak or absent access control** — shared folders rarely have role-based permissions, so anyone with folder access can view or edit sensitive evidence.
- **Manual custody entries** — time-consuming, inconsistently formatted, and easy to miss.
- **No audit trail** — most tools don't log who accessed a record or what they changed, which makes unauthorized access hard to catch after the fact.
- **Poor searchability** — finding one item by case ID, date, or examiner in paper logs or unindexed sheets can eat up significant time.
- **No relational linking** — spreadsheets can't enforce relationships between evidence, cases, suspects, and officers, so duplicate or orphaned records creep in.
- **No lifecycle tracking** — there's no built-in workflow for evidence status, which raises the risk of items getting lost or forgotten in storage.

Why this matters — the practical risks:

1. Legal inadmissibility, when gaps in custody records give the defense grounds to challenge evidence.
2. Undetected tampering, since there's no automated hash verification.
3. Human error from manual entry — typos, missing fields, conflicting records.
4. Scalability problems as case volume grows and spreadsheets stop coping.

5. No disaster recovery for physical logs or unsynced local files (fire, theft, hardware failure).
6. Fragmented reporting, since custody or audit reports have to be manually assembled from scattered sources.
7. Security blind spots, because without authentication logging, insider threats can go unnoticed until real damage is done.

Research Gap

Gap 1 — No practical, database-centric solution

The existing research splits into two camps: purely theoretical work (ontologies, taxonomies, frameworks with no implementation) or heavily blockchain-dependent systems that need infrastructure like Hyperledger Fabric, Ethereum, or Cardano. There's very little middle ground — a practical, relational-database-driven DFEMS that a resource-limited forensic lab or local police department could actually deploy without blockchain overhead.

Gap 2 — The performance-vs-integrity trade-off is still unresolved

Blockchain gives strong immutability but comes with latency, storage overhead, and transaction costs — a limitation the blockchain bibliometric review itself flags as underexplored in real-world deployment. No well-studied model yet combines relational-database efficiency with cryptographic hash verification to get "good enough" tamper-evidence without the full weight of blockchain.

Gap 3 — Access control and audit mechanisms are treated as an afterthought

Most of the systems reviewed focus heavily on logging custody transfers but say little about how role-based access control is actually enforced at the database level for different roles (Investigator, Examiner, Admin, Legal Reviewer).

Gap 4 — Weak modeling of evidence lifecycle and case linkage

Existing studies focus on custody events (who accessed what, and when) but rarely model the full relational structure connecting evidence to cases, suspects, victims, and personnel in a normalized schema — which limits real-world usability for investigators juggling multiple cases at once.

Gap 5 — Limited testing at real operational scale

Proof-of-concept systems like ForensicChain are only tested at small scale. Few studies check search/retrieval performance, reporting automation, or usability under the kind of load a real forensic lab would generate (hundreds or thousands of evidence records).

Gap 6 — Academic research lags behind industry on interoperability

The interoperability patent shows industry has already explored multi-agency data exchange through a dual evidence/custody database design, but this idea barely shows up in academic forensic database research, where most papers still treat DFEMS as single-agency, standalone systems.

Proposed Solution

The proposed system, DFEMS, is built around six modules:

Evidence Intake & Hash Verification Module — registers new evidence with metadata (type, source, date/time, collecting officer), auto-generates a SHA-256/MD5 hash at intake, and re-verifies that hash at every retrieval or transfer, flagging mismatches instantly.

Chain-of-Custody Module — automatically logs every access, transfer, or analysis event (who, when, what action, and why), using an append-only log table with no edit or delete privileges for any role, so the trail stays tamper-evident. Generates court-ready custody reports on demand.

Case & Personnel Management Module — links evidence to Case ID, Suspects, Victims, and Investigating Officers, and enforces relational integrity through foreign keys so orphaned or duplicate records can't happen.

Role-Based Access Control (RBAC) Module — predefined roles (Admin, Investigator, Forensic Examiner, Legal/Court Reviewer), each with scoped view/edit/export permissions enforced at both the database and application layer, plus logging of every login attempt, successful or failed.

Evidence Lifecycle & Status Tracking Module — tracks status through Collected → In Storage → Under Analysis → Presented in Court → Archived/Disposed, with alerts for anything stuck in one stage longer than expected.

Search & Reporting Module — indexed search by Case ID, evidence type, examiner, date, or status, plus one-click generation of custody logs, audit trails, and case summary reports in print-ready PDF form.

Together, these modules close the gaps identified above: they give the system the practical, relational-database foundation that Gap 1 says is missing, pair database performance with hash-based integrity to address Gap 2, bake RBAC and audit logging directly into the data layer for Gap 3, and model the full case-evidence-personnel relationship for Gap 4.

System Requirements Gathering

Stakeholders

Stakeholder	Role in the System	Interest / Concern
Police Department / Law Enforcement Agency	Primary evidence collectors and case owners	Reliable evidence tracking, legal defensibility
Forensic Lab / Examiners	Analyze digital evidence	Fast, secure access to assigned evidence
Court / Judiciary	Consumers of final evidence & custody	

	reports	Admissibility, tamper-proof documentation
Legal Prosecutors / Defense Counsel	Review evidence and custody trail for case proceedings	Transparent, verifiable audit trail
System Administrator	Manages users, roles, and system configuration	System uptime, security, data backup
IT / Cybersecurity Department	Maintains infrastructure and security compliance	Data protection, breach prevention
Evidence Custodian / Store Officer	Manages physical/digital evidence storage	Accurate lifecycle status tracking
Auditors / Compliance Officers	Periodic system and process audits	Complete audit logs, compliance reporting

System Users (Roles & Access Levels)

User Role	**Description**	**Access Level**
Admin	Manages users, roles, system settings, full data access	Full CRUD on all modules
Investigator	Registers cases, links suspects/victims, requests evidence	Create/View evidence & case data; no delete

Forensic Examiner	Analyzes evidence, updates analysis status/findings	View assigned evidence; add analysis notes; cannot delete records
Evidence Custodian	Manages evidence intake, storage, and status updates	Create/update evidence records & status; view custody logs
Legal/Court Reviewer	Reviews finalized evidence & custody reports for legal proceedings	Read-only access to case evidence & custody logs
Auditor	Reviews system logs and access history	Read-only access to audit trail & custody logs (no case data edits)

Data Requirements (Entities to be Captured)

Entity	**Key Attributes**
Evidence	Evidence ID, Type, Description, Source, Hash Value, Status, Date Collected
Case	Case ID, Case Title, Crime Type, Date Opened, Status
Suspect	Suspect ID, Name, Details, Linked Case(s)

Officer/Personnel	Officer ID, Name, Role, Badge/Department
Custody Log	Log ID, Evidence ID, Action, Performed By, Timestamp, Reason
User Account	User ID, Username, Password (hashed), Role, Status
Audit Log	Log ID, User ID, Action, Timestamp, IP/Device (optional)

Functional and Non-Functional Requirements

Functional Requirements

ID	Requirement	Description
FR1	Evidence Registration	Authorized users can register new evidence with metadata (type, source, date, collecting officer)
FR2	Hash Generation	System auto-generates a SHA-256/MD5 hash on evidence intake
FR3	Hash Re-verification	System re-verifies the hash at every access/transfer and flags mismatches

FR4	Chain-of-Custody Logging	System automatically logs every access, transfer, or status change (who, when, what, why)
FR5	Case-Evidence Linking	Evidence can be linked to Case ID, Suspects, Victims, and Officers
FR6	Role-Based Access Control	Actions and data visibility are restricted based on the user's assigned role
FR7	Authentication	Secure login required (username/password, hashed credentials)
FR8	Status/Lifecycle Tracking	System tracks evidence status: Collected → Stored → Analyzed → Presented → Archived/Disposed
FR9	Search & Filter	Users can search by case ID, evidence type, date, examiner, or status

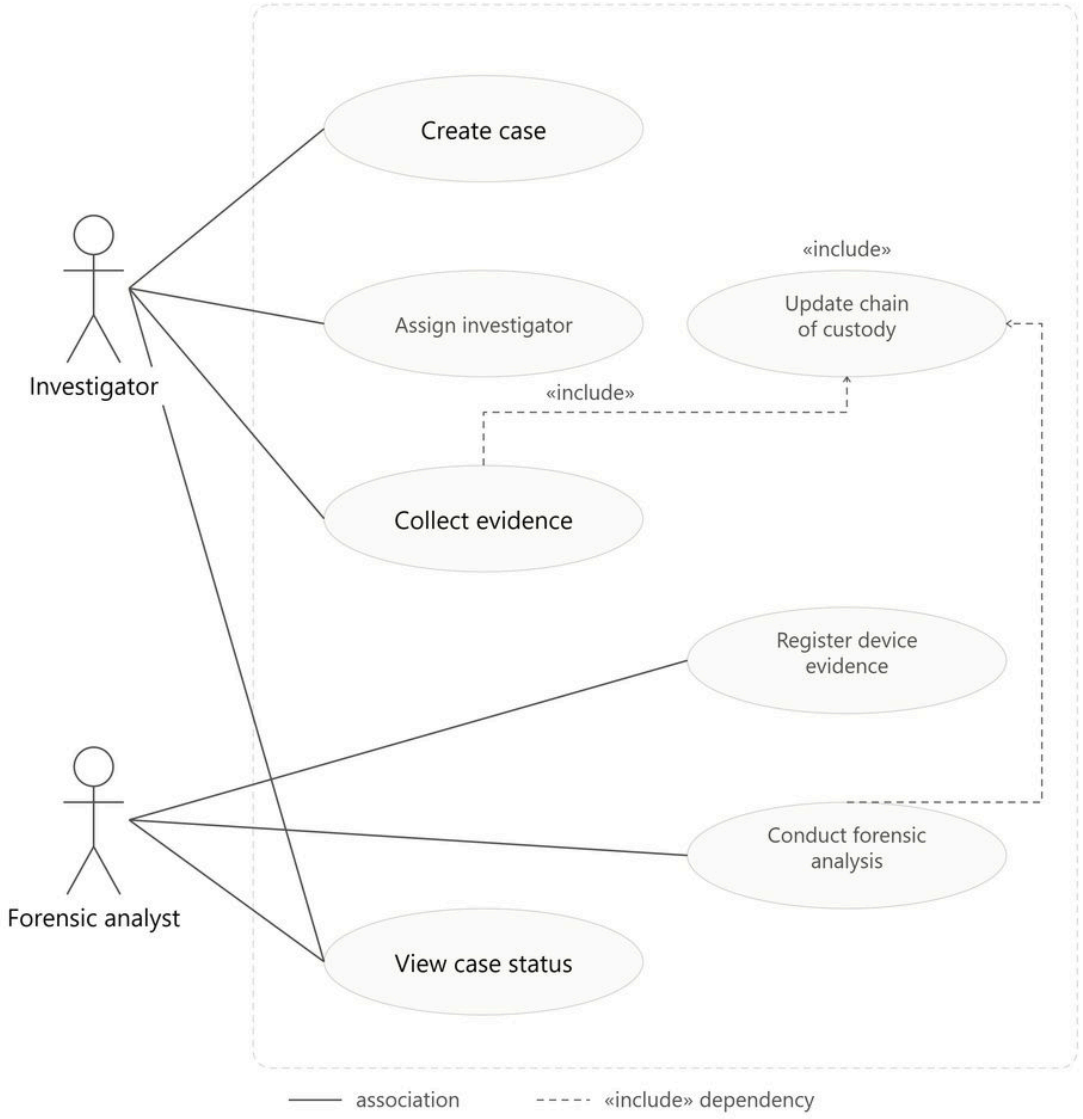
FR10	Report Generation	System generates custody logs, audit trails, and case summary reports (PDF/print)
FR11	Audit Logging	All login attempts (successful/failed) and data access events are logged
FR12	Data Validation	Mandatory fields and referential integrity (foreign key constraints) are enforced

Non-Functional Requirements

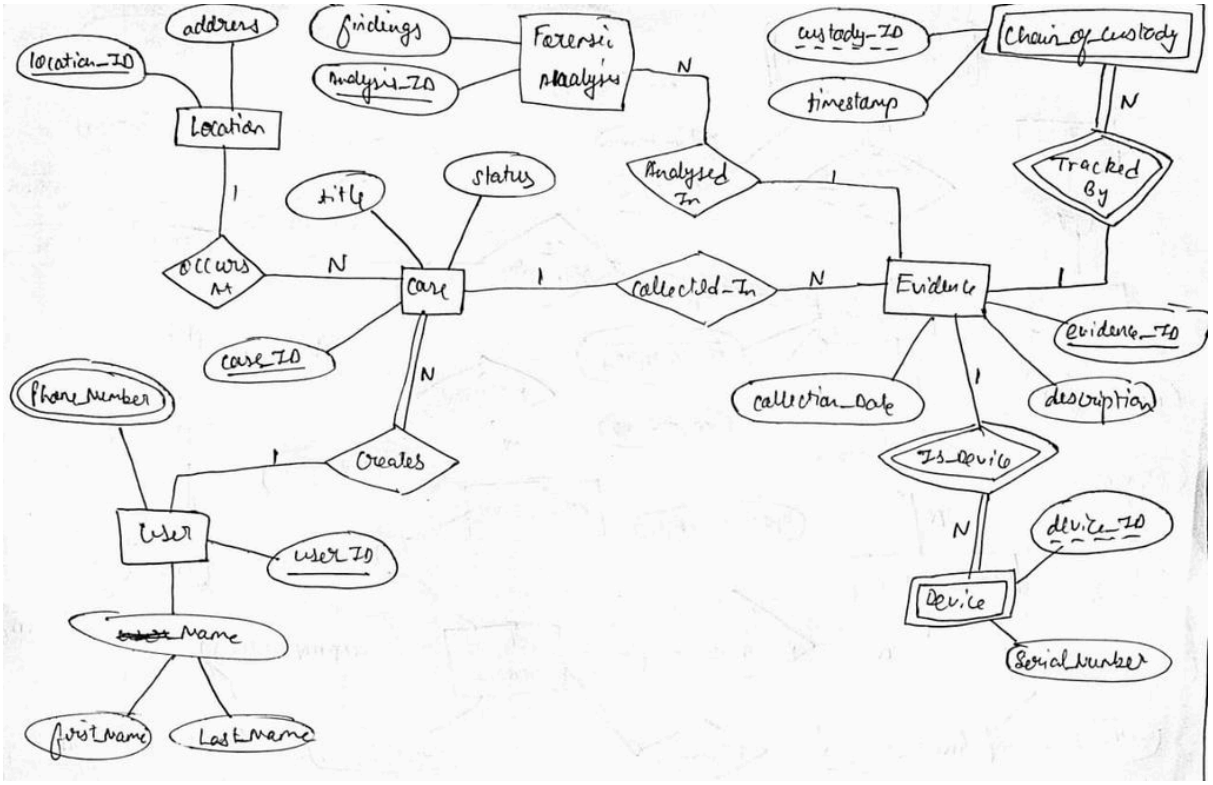
ID	Requirement	Description	Target
NFR1	Performance	Evidence search/retrieval response time	Under 10 seconds
NFR2	Security	Data encryption at rest and in transit	AES-256 / TLS 1.2+
NFR3	Availability	System uptime for operational reliability	≥ 99%

NFR4	Scalability	Support growing case/evidence volume without performance loss	Up to 100,000+ records
NFR5	Data Integrity	Hash verification accuracy	100%
NFR6	Auditability	All access/actions logged and immutable	100% action coverage
NFR7	Usability	Intuitive UI for non-technical investigators	Minimal training required
NFR8	Backup & Recovery	Regular automated backups with recovery capability	Daily backup, under 1 hr recovery
NFR9	Compliance	Aligns with digital evidence handling standards (e.g., ISO/IEC 27037)	Full compliance

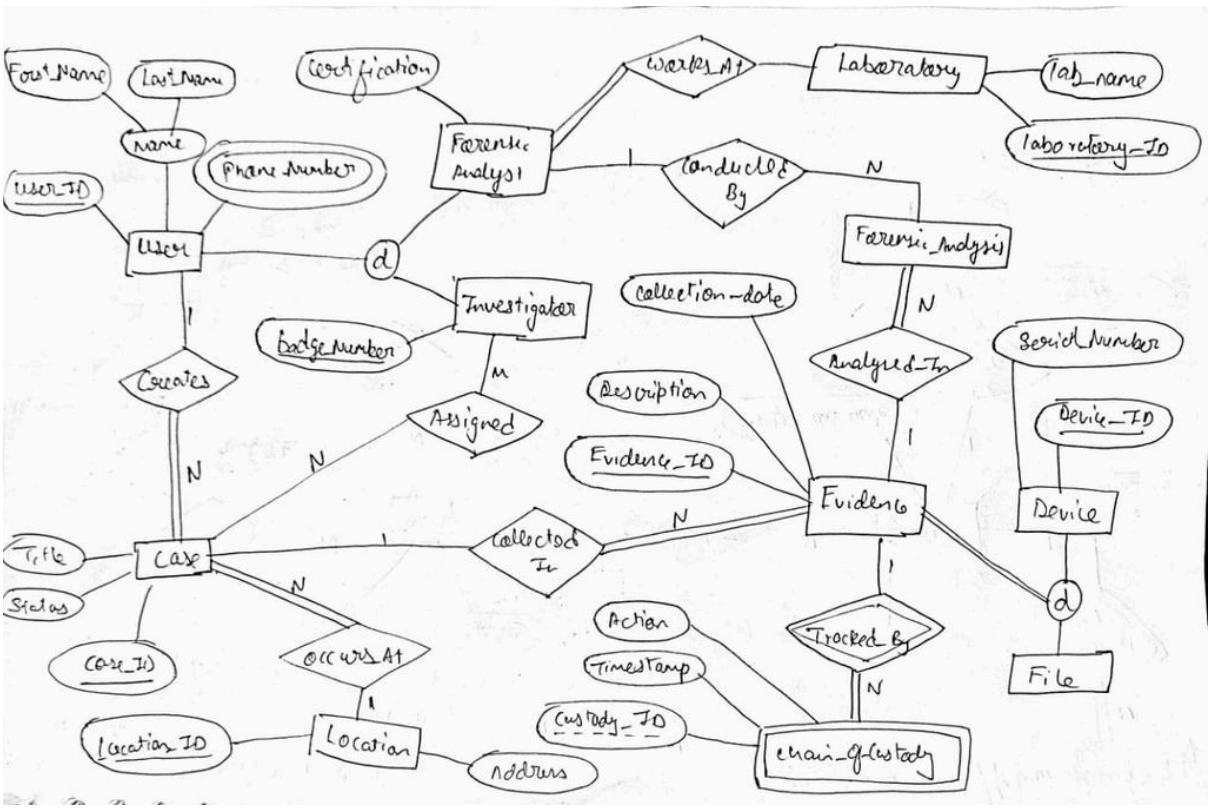
Use Case Diagram


Create case
Investigator
«include»
Assign investigator
Update chain of custody
«include»
Collect evidence
Register device evidence
Forensic analyst
Conduct forensic analysis
View case status
association
«include» dependency

ER Diagram


address
findings
Forensic analysis
custody_ID
Chain_of_custody
location_ID
Location
analysis_ID
timestamp
N
Tracked By
Analysed In
title
status
occurs At
N
Case
1
collected_In
N
Evidence
case_ID
evidence_ID
Phone_Number
collection_date
description
Creates
Is_device
User
user_ID
device_ID
Name
Device
first_Name
Last_Name
Serial_Number

EER Diagram


First_Name
Last_Name
Name
Phone_Number
User_ID
User
Creates
Certification
Forensic Analyst
Works_At
Laboratory
Lab_name
laboratory_ID
Conducted By
Forensic_Analysis
d
Investigator
Badge_Number
Assigned
collection_date
Description
Evidence_ID
Analysed_In
Serial_Number
Device_ID
Device
Evidence
Title
Status
Case
collected In
Tracked_By
Action
Timestamp
Custody_ID
chain_of_custody
File
Case_ID
occurs At
Location_ID
Location
Address

Data Dictionary

1. USER

Attribute	Data Type	Key	Description
User_ID	INT	PK	Unique identifier for a system user
First_Name	VARCHAR(50)		User's first name
Last_Name	VARCHAR(50)		User's last name
Phone_Number	VARCHAR(15)		User's contact phone number

1a. INVESTIGATOR

Attribute	Data Type	Key	Description
User_ID	INT	PK,FK->USER	Identifies the user acting as an investigator
Badge_Number	VARCHAR(20)		Investigator's badge/ID number

1b. FORENSIC ANALYST

Attribute	Data Type	Key	Description
User_ID	INT	PK,FK->User	Identifies the user acting as a forensic analyst
Certification	VARCHAR(50)		Analyst's professional certification reference
Laboratory_ID	INT	FK->Laboratory	Lab the analyst works at

2. CASE

Attribute	Data Type	Key	Description
Case_ID	INT	PK	Unique identifier for a case
Title	VARCHAR(100)		Title/Name of the case
Status	VARCHAR(20)		Current status of the case
Assigned_To	INT	FK->Investigator	Investigator assigned to the case

Created_By	INT	FK->User	User who created the case
Location_ID	INT	FK->Location	location where the case occured

3. LOCATION

Attribute	Data Type	Key	Description
Location_ID	INT	PK	Unique identifier for a location
Address	VARCHAR (150)		physical address of the location

4. EVIDENCE

Attribute	Data Type	Key	Description
Evidence_ID	INT	PK	Unique identifier for a piece of evidence
Description	VARCHAR(255)		Description of the evidence item

Collection_date	DATE		Date the evidence was collected
Case_ID	INT	FK->Case	Case the evidence belongs to

4a.DEVICE

Attribute	Data Type	Key	Description
Device_ID	INT	PK	Unique identifier for a device
Serial_Number	VARCHAR(50)		Manufacturer serial number of the device
Evidence_ID	INT	FK->Evidence	Evidence record this device corresponds to

4b.FILE

Attribute	Data Type	Key	Description

File_ID	INT	PK	Unique identifier for a digital file record
Evidence_ID	INT	FK->Evidence	Evidence record this file corresponds to

5. CHAIN OF CUSTODY

Attribute	Data Type	Key	Description
Custody_ID	INT	PK	Unique identifier for a custody log entry
Timestamp	DATETIME		Date/time the custody event occurred
Action	VARCHAR(50)		Action performed
Evidence_ID	INT	FK->Evidence	Evidence item the custody record tracks

6. FORENSIC_ANALYSIS

Attribute	Data Type	Key	Description
Analysis_ID	INT	PK	Unique identifier for a forensic analysis record
Evidence_ID	INT	FK->Evidence	Evidence item that was analyzed
Analyst_ID	INT	FK->Forensic Analyst	Analyst who conducted the analysis

7. LABORATORY

Attribute	Data Type	Key	Description
Laboratory_ID	INT	PK	Unique identifier for a laboratory
Lab_Name	VARCHAR(100)		Name of the laboratory

Relational Schema

USER(User_ID,First_Name,Last_Name,Phone_Number,Laboratory_ID)

PK:User_ID

INVESTIGATOR(User_ID,Certification,Laboratory_ID)

PK:User_ID

FK:User_ID->User(User_ID)

FORENSIC_ANALYST(User_ID,Certification,Laboratory_ID)

PK:User_ID

FK:User_ID->User(User_ID)

FK:Laboratory_ID->Laboratory(Laboratory_ID)

LABORATORY(Laboratory_ID,Lab_Name)

PK:Laboratory_ID

LOCATION(Location_ID,Address)

PK:Location_ID

CASE(Case_ID,Title,Status,Created_By,Assigned_To,Location_ID)

PK:Case_ID

FK:Created_By->User(User_ID)

FK:Assigned_To->User(User_ID)

FK:Location_ID->Location(Location_ID)

ASSIGNED(User_ID,Case_ID)

PK:(User_ID,Case_ID)

FK:User_ID->Investigator(User_ID)

FK:Case_ID->Case(Case_ID)

EVIDENCE(Evidence_ID,Description,Collection_Date,Case_ID)

PK:Evidence_ID

FK:Case_ID->Case(Case_ID)

DEVICE(Device_ID,Serial_Number,Evidence_ID)

PK:Device_ID

FK:Evidence_ID->Evidence(Evidence_ID)

FILE(File_ID,Evidence_ID)

PK:File_ID

FK:Evidence_ID->Evidence(Evidence_ID)

CHAIN_OF_CUSTODY(Custody_ID,Timestamp,Action,Evidence_ID)

PK:Custody_ID

FK:Evidence_ID->Evidence(Evidence_ID)

Forensic_Analysis(Analysis_ID,Findings,Timestamp,Certification,Evidence_ID,Analyst_ID)

PK:Analysis_ID

FK:Evidence_ID->Evidence(Evidence_ID)

FK:Analyst_ID->Forensic_Analyst(User_ID)

FUNCTIONAL DEPENDENCIES

1. USER

FD	BUSINESS RULE
User_ID-> First_Name,Last_Name,Phone_No	User_ID uniquely identifies a user
Phone_Number->User_ID	Each phone number belongs to one user

1a.INVESTIGATOR

FD	BUSSINESS RULE
User_ID->Badge_Number	User uniquely determines their badge number
Badge_Number->User_ID	badge numbers are unique so no two investigators share a badge

1b.FORENSIC_ANLAYST

FD	BUSSINESS RULE

User_ID->Certification,Laboratory_ID	User_ID uniquely determines their certification and the lab they work at

2. LABORATORY

FD	BUSINESS RULE
Laboratory_ID->Lab_Name	Laboratory_ID uniquely identifies a lab
Lab_Name->Laboratory_ID	Lab names are unique

3. LOCATION

FD	BUSSINESS RULE
Location_ID->Address	Location uniquely identifies a location

4. CASE

FD	BUSSINESS RULE

5. ASSIGNED

FD	BUSSINESS RULE
User_ID,Case_ID->(no non-key attributes)	The combination of investigator and case is unique so an investigator is assigned to a case at most once

6. EVIDENCE

FD	BUSSINESS RULE
Evidence_ID->Description,Collection_date,Case_ID	Evidence_ID,uniquely indentifies an evidence item

6a.DEVICE

FD	BUSSINESS RULE
Device_ID->Serial_Number,Evidence_ID	Device_ID uniquely identifies a device and the evidence record it corresponds to
Serial_Number->Device_ID	Manufacturer serial numbers are unique
Evidence_ID->Device_ID	Each evidence record maps to at most one device

6b.FILE

FD	BUSSINESS RULE
File_ID->Evidence_ID	File_ID uniquely identifies a digital file record and the evidence record it corresponds to
Evidence_ID->File_ID	Each evidence record maps to at most one file

7. CHAIN_OF_CUSTODY

FD	BUSSINESS RULE
Custody_ID-> Timestamp,Action,Evidence_ID	Custody_ID uniquely identifies a custody log entry

8. FORENSIC_ANALYSIS

FD	BUSSINESS RULE
Analysis_ID-> Evidence_ID,Analyst_ID	Analysis_ID uniquely identifies an analysis record

NORMALIZATION

First Normal Form(1NF)

Relation	Atomic Attribute	Repeating groups	1NF
User	Yes	None	Yes
Investigator	Yes	None	Yes
Forensic_Analyst	Yes	None	Yes
Location	Yes	None	Yes
Case	Yes	None	Yes
Assigned	Yes	None	Yes
Evidence	Yes	None	Yes
Device	Yes	None	Yes
File	Yes	None	Yes
Chain_of_Custody	Yes	None	Yes
Forensic_Analysis	Yes	None	Yes

Result:All 12 relations are already in 1NF , every attribute holds a single atomic value, and no relation has multi valued .

Second Normal Form(2NF)

Relation	Primary Key	Composite	Partial Dependenc y	2NF
User	User_ID	No	No	Yes
Investigator	User_ID	No	No	Yes
Forensic_Anal yst	User_ID	No	No	Yes
Location	Location_ID	No	No	Yes
Case	Case_ID	No	No	Yes
Assigned	(User_ID,Cas e_ID)	Yes	No	Yes
Device	Device_ID	No	No	Yes
Evidence	Evidence_ID	No	No	Yes
File	File_ID	No	No	Yes
Chain_of_Cu stody	Custody_ID	No	No	Yes
Forensic_An alysis	Analysis_ID	No	No	Yes

Result:11 of the 12 relations use single attribute PK,so partial dependency is structurally impossible .Assigned is the one composite key relation , but it has

no non key attribute at all, so there is nothing that could depend on only part of the key.All 12 relations are in 2NF.

Third Normal Form(3NF)

1. User:First_Name,Last_Name,Phone_Number,each depends only on User_ID.

1a.Investigator:Badge_Number depends only on User_ID.

1b.Forensic_Analyst:Certification,laboraotry_ID each depends only User_ID not on each other.

2. Laboratory:Lab_name depends only on a Laboratory_ID.

3. Location:Address depends only on Location_ID.

4. Case:Title,Status,Created_By,Assigned_To,Location_ID,each depends only on Case_ID.

5. Assigned:No non key attributes.

6. Evidence:description,Collection_Date,Case_ID,each depends only on Evidence_ID,not on each other.

6a.Device:Serial_Number,Evidence_ID each depends only on Device_ID.

6b.File:Evidence_ID depends only on File_ID.

7. Chain_of_Custody:Timestamp,Action,Evidence_ID,each depends only on Custody_ID.

8. Forensic_Analysis:Evidence_ID,Analyst_ID,each depends on Analyst_ID.

Result:No transitive dependencies found in any relation. All 12 relations are in 3NF , no decomposition need at this step.

Boyce-Codd Normal Form(BCNF)

Relation	Determinants(X in X->Y)	Is X is Super Key?	BCNF

User	User_ID,Phone_Number	All two are candidate key	Yes
Investigator	User_ID,Badge_Number	Both candidate key	Yes
Forensic_Analyst	User_ID	candidate key	Yes
Laboratory	Laboratory_ID,Lab_Name	Both candidate key	Yes
Location	Location_ID	Candidate Key	Yes

Assigned	(User_ID,Case_ID)	Composite candidate key	Yes
Device	Device_ID,Serial_Number	Candidate key	Yes
Evidence	Evidence_ID	Candidate key	Yes
File	File_ID,Evidence_ID	Both candidate keys	Yes
Chain_of_Custody	Custody_ID	Candidate Key	Yes
Forensic_Analysis	Analysis_ID	Candidate key	Yes

Result:Every determinant in every relation is a candidate key.All 12 relation satisfy BCNF, no further decomposition is required.